What a Data Scientist Actually Does

Lesson 1 · the job, the titles and the week

In this lesson → ① Tuesday question ② 4 titles ③ Where time goes ④ Skill stack ⑤ The loop ⑥ Myths ⑦ Interview Q&A

Module dataset: ChurnLens, a deliberately messy telecom CSV we clean, explore and model across Module 1.

1 Four titles, one team

> A **data scientist** turns data into decisions, using statistics and code — and spends most of the week on the data, not the model.

Role	Answers	Main tools
Data engineer	is the data there, on time, correct?	SQL, Spark, Airflow, warehouses
Data analyst	what happened?	SQL, dashboards, Excel
Data scientist	why, and what next?	Python, pandas, stats, sklearn
ML engineer	does it serve real traffic?	PyTorch, Docker, APIs, MLOps

Small team
One person often plays all four roles. The titles matter less than the pipeline: **raw data → answers → decisions → products.**

2 Where the week actually goes



the model is the small part

→ ~46% finding, joining and cleaning data
→ ~22% analysis, charts, sanity checks
→ ~18% talking to people: framing the question, presenting the answer
→ ~14% modelling

Reality
If you enjoy the cleaning and the conversations, you will enjoy this job.

3 The skill stack

Skill	Why it matters	Good enough looks like
SQL	the data lives in a database; every interview tests it	joins, GROUP BY, window functions
Python + pandas	everything SQL cannot do	load, clean, merge, group, plot
Statistics	knowing when a difference is real	distributions, CIs, hypothesis tests
Machine learning	prediction, and explaining trade-offs	regression, trees, metrics, validation
Communication	decides whether the work changes anything	one slide, one number, one recommendation
A little engineering	getting a model out of the notebook	git, an API, Docker basics

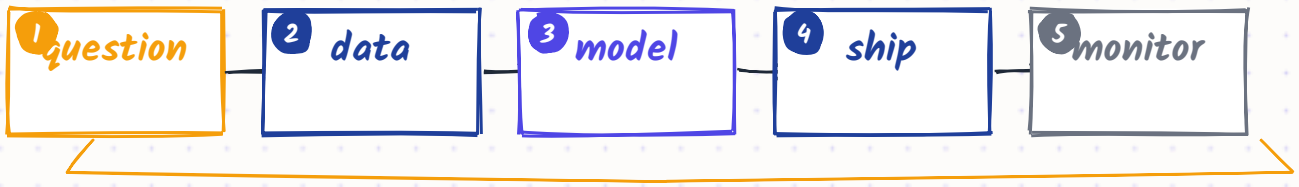
Hired as an analyst

- ✓ SQL
- ✓ charts
- ✓ clear writing

Hired as a data scientist

- ✓ the above
- ✓ statistics
- ✓ ML + validation

4 A project, end to end



- **Question** — who will act on the answer, and how?
- **Data** — find it, check it, clean it, document what you changed
- **Model or analysis** — the simplest thing that answers the question
- **Ship** — a dashboard, a report, or an endpoint
- **Monitor** — the world changes; models go stale quietly

Loop
What you learn at step 5 sends you back to step 1. That is the job, repeated.

5 What you do NOT need

Myth	Reality
You need a PhD	most posts ask for a portfolio; a PhD helps for research roles
You must be a maths genius	be comfortable with averages, percentages, probability
You need big-company data	a clean public-dataset project beats a vague private one
You need the newest model	boring methods, explained clearly, win interviews

6 Interview corner

Q. What is data science, in one line?

A. Turning data into decisions using statistics and code.

Q. Analyst vs scientist vs ML engineer?

A. Analyst: what happened. Scientist: why and what next. ML engineer: makes it serve traffic.

Q. Signups dropped 12%. What do you do?

A. Check tracking first → split by segment, date, platform → compare like for like → then look for a cause. Never start with a model.

Q. Which part of a project takes longest?

A. Finding and cleaning the data — roughly half the time. Say this; interviewers are checking you have done real work.

⚡ Revise in 60 seconds

- → 4 roles: engineer, analyst, scientist, ML engineer
- → ~50% of the week is data, not models
- → stack: SQL · Python · stats · ML · communication
- → 5 steps: question → data → model → ship → monitor
- → metric drop: tracking → segments → like-for-like → cause
- → portfolio beats a PhD for most roles

Next lesson → DS-02
The **data science workflow** end to end — and the 3 places projects die
Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**